

Improve and Challenge

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can repair a specific observed failure.

I can check that earlier behavior still works.

I can keep a truthful change and failure record.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Test Diverse Users. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Evidence and Sources. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 35: Improve and Challenge

A controlled improvement changes one design choice and reruns the same tests.



THINKER MISSION

Compare before/after scores and explain the tradeoff.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why rerun old passing cases after a fix?

Readiness for Testing functions and interfaces

Why can testing only `valid_item` miss a pipeline bug?

My readiness explanation

This week's end goal

By the end, I can make a focused repair and challenge it with regression and order-invariance tests.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use improve and challenge to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional chooser wrongly excludes an activity at the exact time limit because it uses $\text{minutes} < \text{limit}$. The requirement says at most, so equality must qualify. Repair that condition, then test below, at and above the boundary plus earlier tie and invalid-input behavior.

Apply the connected idea: Testing functions and interfaces

Test the chooser at two scales. A unit test checks one validator or score calculation. An integration test runs validation through selection and reporting. Write expected outcomes from the specification, not by calling the same function twice.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Compare before/after scores and explain the tradeoff.



PROCESS

A controlled improvement changes one design choice and reruns the same tests.



OUT

OUTPUT

Compare before/after scores and explain the tradeoff.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def eligible(minutes, limit):  
    return minutes <= limit  
  
for minutes, expected in [(9, True), (10, True), (11, False)]:  
    actual = eligible(minutes, 10)  
    assert actual == expected  
    print(minutes, actual)
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Before fix

Evidence: $10 < 10$ is False

Meaning: Violates at most rule

Step or case: After fix

Evidence: $10 \leq 10$ is True

Meaning: Boundary accepted

Step or case: Neighbor check

Evidence: $11 \leq 10$ is False

Meaning: Over-limit still excluded

Read each row from left to right and connect the evidence to the stated rule.

The repaired predicate prints 9 True, 10 True and 11 False. It assumes validated numeric durations; it is not a replacement for the schema validator. A focused change is easier to explain and evaluate than unrelated edits made at the same time.

Repair the rule, then keep the evidence

At most includes equality. For a limit of 10, the durations 9 and 10 qualify while 11 does not. The strict comparison `minutes < limit` wrongly rejects 10; changing it to `minutes <= limit` repairs that boundary. Check below, at and above the boundary, then rerun earlier tie and invalid-input cases. Keep the original failure record so another person can understand what changed and which evidence supports the repair.

Apply the connected idea: Testing functions and interfaces

```
def score(item):
    return 2 * item["fun"] + item["learning"]

cases = [( {"fun": 0, "learning": 0}, 0),
          ( {"fun": 3, "learning": 2}, 8),
          ( {"fun": 5, "learning": 5}, 15)]

for item, expected in cases:
    assert score(item) == expected

print("3 score cases passed")
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Unit

Evidence: Score 3 and 2

Meaning: Expect 8

Step or case: Integration

Evidence: Invalid high-score row

Meaning: Must never win

Step or case: Report

Evidence: No eligible rows

Meaning: Explain no suggestion

Read each row from left to right and connect the evidence to the stated rule.

The code prints its success message only if all assertions pass. It checks the scoring component but says nothing about duration filtering, malformed records, ties or reporting. Add focused cases for those requirements. Use the production functions from earlier weeks rather than rewriting a second copy just for tests.

Test the connections as well as the pieces

A score unit test asks whether the scoring function calculates the specified number. An integration test asks whether validation, filtering, scoring and reporting work together. A correct score does not establish that invalid records were rejected or that a tie was resolved correctly. Predict the full report from the input records before running. Check its counts and reasons as well as its selected name, and repeat a tie case with reversed input order.

Connect the picture and the explanation

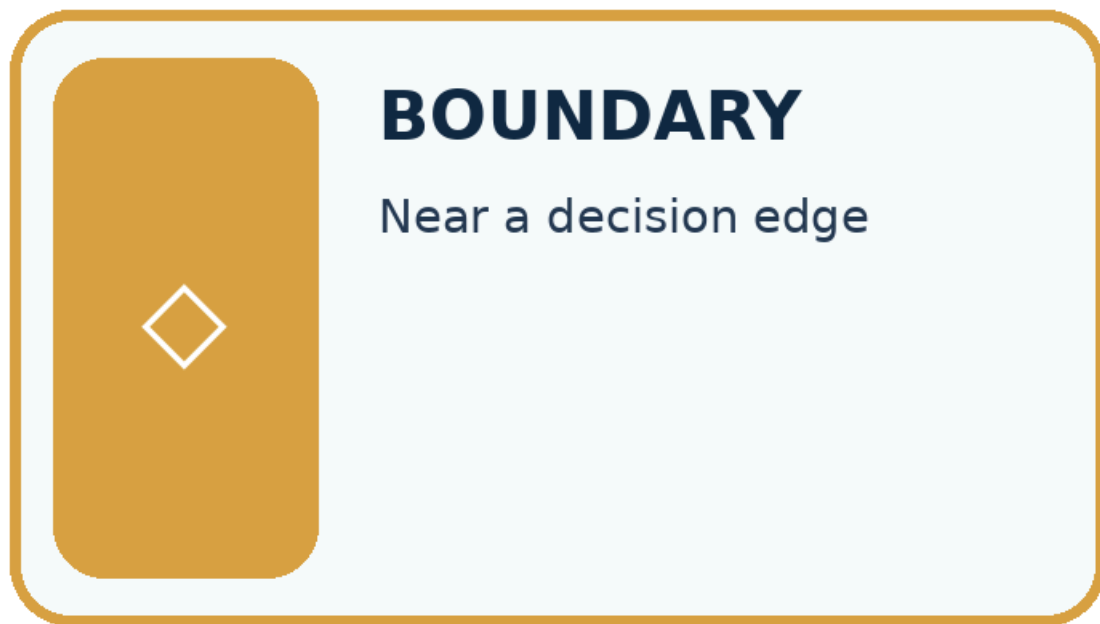
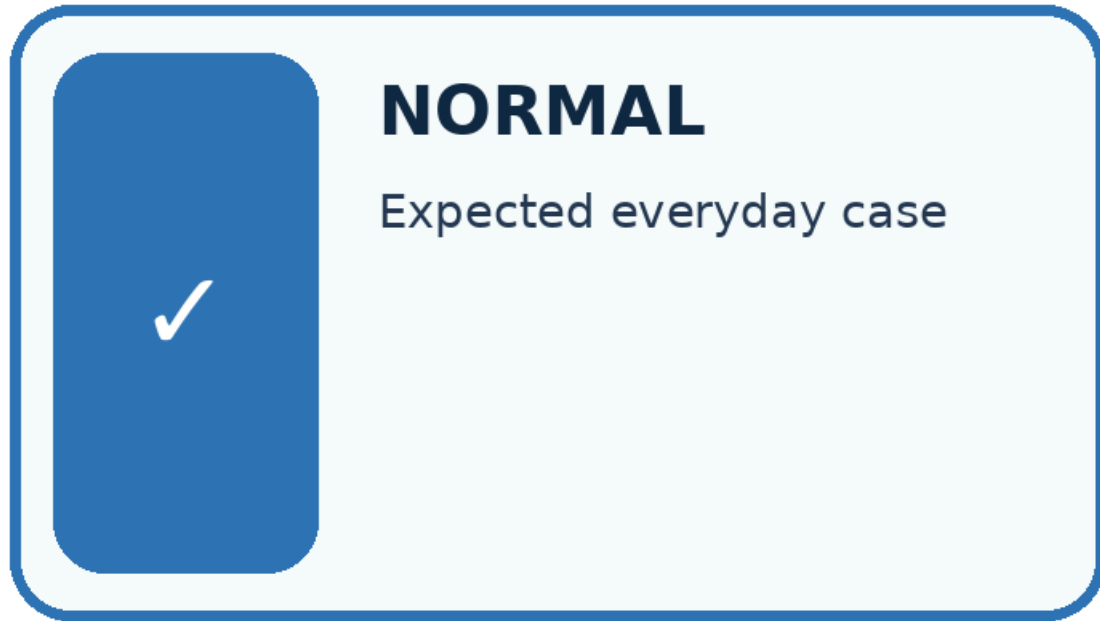
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

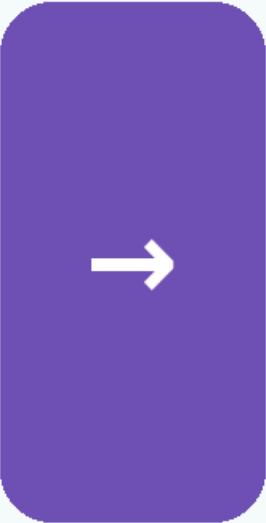


NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: An alternative patch simply increases the limit to 101.

Prediction: C is wrongly admitted as well as A.

Reason: Changing the maximum alters the contract instead of repairing its comparison.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

CONTROLLED IMPROVEMENT
one planned change tied to evidence

2

BEFORE/AFTER

results from the same test
before and after

3

TRADEOFF

benefit paired with a cost

4

REGRESSION

previously working behavior
becomes worse

Word · My own example · Not this / boundary

Word: controlled improvement

My own example: A revision whose effect is checked under comparable conditions rather than assumed from a changed result.

Not this / boundary: _____

Word: before/after

My own example: Results recorded prior to and following a change; other changes can affect the comparison.

Not this / boundary: _____

Word: tradeoff

My own example: A situation where improving one desired property can make another harder to achieve.

Not this / boundary: _____

Word: regression

My own example: In testing, a new failure introduced by a change; in modelling, prediction of a numeric target. Use the meaning indicated by the lesson.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Regression suite

Meaning: Earlier cases rerun after a change

Word: Change log

Meaning: A record of what changed and why

Word: Root cause

Meaning: The underlying reason for a failure

Word: Residual risk

Meaning: A limitation or possible failure remaining after a repair

Terms for Testing functions and interfaces

Term · Meaning

Term: Unit test

Meaning: A check of one small component

Term: Integration test

Meaning: A check of connected components

Term: Fixture

Meaning: A controlled input used in a test

Term: Oracle

Meaning: An independent source of the expected result

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 35 • WORKED EVIDENCE

- 1 Before: 6/10 test profiles pass.
- 2 Error cluster: difficult books rank too high for emerging readers.
- 3 Change only difficulty weight from 1 to 3.
- 4 After: 8/10 pass; one adventure-focused profile ranks worse—record the tradeoff.

- 1. Before: 6/10 test profiles pass.
- 2. Error cluster: difficult books rank too high for emerging readers.
- 3. Change only difficulty weight from 1 to 3.
- 4. After: 8/10 pass; one adventure-focused profile ranks worse—record the tradeoff.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Would changing the condition to `minutes <= limit + 1` correctly repair the boundary? Explain with a revealing input.

Apply the connected idea: Testing functions and interfaces

A test computes `expected = score(item)` then compares `score(item)` to `expected`. Why is it weak?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Improve and Challenge

A fictional recommender excludes books at an intended including equality page limit.

The facts we will use

Contract: `pages <= 100`; highest interest; ties choose smaller ID. Old code uses `pages < 100`.

Rows A `pages100 interest5`; B `pages90 interest3`; C `pages101 interest5`.

Improve and Challenge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify cause

A wrongly excluded at equality, so old code selects B.



2. Focused repair

Replace `<100` with `<=100` without changing ranking or type checks.



3. Expected revised result

A becomes eligible and wins; C remains excluded.



4. Challenge repair

Rerun 99,100,101; invalid pages; tied scores; reversed row order.

CHANGE ONE THING

An alternative patch simply increases the limit to 101.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Identify cause

What happens and why: A wrongly excluded at equality, so old code selects B.

Step: 2. Focused repair

What happens and why: Replace `<100` with `<=100` without changing ranking or type checks.

Step: 3. Expected revised result

What happens and why: A becomes eligible and wins; C remains excluded.

Step: 4. Challenge repair

What happens and why: Rerun 99,100,101; invalid pages; tied scores; reversed row order.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. Regression tests check surrounding behavior that should remain unchanged. A tie case and reversed input order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

What this example does not establish

Passing a small regression suite does not prove all behavior unchanged; record the suite's actual coverage.

Read the labelled picture: Why are tests at both 100 and 101 needed?

Change one thing: An alternative patch simply increases the limit to 101. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.

!

COMPARE • DIAGNOSE

Weak idea: If the total score improves, no old behavior needs retesting. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

A clean regression run does not erase the earlier failure. Keep before/after evidence and remaining limitations. If a fix was informed by reserved evaluation cases, report that contamination and obtain new independent cases.

Record the bug, change, evidence and one remaining limit without claiming all bugs are gone.

Apply the connected idea: Testing functions and interfaces

Test results establish only the executed cases on the tested version. Retain failures and version information. Use explicit expected rejection behavior for invalid data instead of assuming every exception is acceptable.

What expected result should an empty-input integration case specify?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Changing the expected boundary result to False would make the test pass while violating the requirement. Keep the requirement and independently calculated expectation; repair the implementation. If requirements truly change, document that as a policy change.

Apply the connected idea: Testing functions and interfaces

A test suite that only checks the winner name can miss wrong rejection counts. Check the full expected report: supplied, accepted, rejected, eligible, selected and reason. Avoid relying on dictionary display order when the contract concerns values.

Practise with less help: Test neighbors

Use workbook W2 for the connected example “Improving without hiding failures”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Improving without hiding failures. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Testing functions and interfaces

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Why are tests at both 100 and 101 needed?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Choose one evidence-backed change, freeze the test set, predict effects, rerun every case, compare 6/10 to 8/10, and inspect regressions.

Choose one evidence-backed change, freeze the test set, predict effects, rerun every case, compare 6/10 to 8/10, and inspect regressions.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

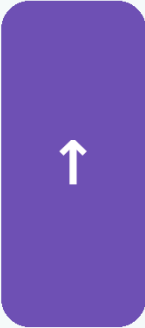
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Publish the cause, one change, fixed test set, before/after evidence, and remaining limitation.

THIS WEEK'S CHECKPOINT Publish the cause, one change, fixed test set, before/after evidence, and remaining limitation.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

For each recommender revision, retain cause, code change, targeted test, regression results and unresolved issues.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Compare before/after scores and explain the tradeoff.

CHALLENGE Compare before/after scores and explain the tradeoff.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 35 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A repair raises success from 7/10 to 9/10 on the same validation set.

The question

What belongs in the improvement claim?

Worked explanation

Confirm fixed criteria, inspect remaining and introduced failures, then test the chosen version on fresh final cases.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Improve with comparable evidence

Evidence or stage · Value or result

Evidence or stage: Before

Value or result: Save results and failures

Evidence or stage: Revision

Value or result: Change one identified cause

Evidence or stage: Retest

Value or result: Use the same development cases

Evidence or stage: Final check

Value or result: Reserve fresh cases

A repair raises success from 7/10 to 9/10 on the same validation set.

What belongs in the improvement claim?

Confirm fixed criteria, inspect remaining and introduced failures, then test the chosen version on fresh final cases.

14. Try a new case independently

Independent case: Repair independently

Write the eligibility predicate for at most 12 minutes and a regression set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 35 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

After changing a scoring weight, the system fixes one request but breaks two previously correct ones. What should the improvement record say?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Publish the cause, one change, fixed test set, before/after evidence, and remaining limitation.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional recommender excludes books at an intended including equality page limit.

By the end, I can · How I will show it

By the end, I can: Repair a specific observed failure

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Check that earlier behavior still works

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Keep a truthful change and failure record

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Repair a specific observed failure

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender excludes books at an intended including equality page limit.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Rerun 99,100,101; invalid pages; tied scores; reversed row order.

- 1. Identify cause
- 2. Expected revised result
- 3. Challenge repair
- 4. Focused repair

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Challenge repair Rerun 99,100,101; invalid pages; tied scores; reversed row order. A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. Regression tests check surrounding behavior that should remain unchanged. A tie case and reversed input order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Practice 2: explain the deciding evidence

Changed-case practice: An alternative patch simply increases the limit to 101. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. C is wrongly admitted as well as A.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: C is wrongly admitted as well as A. Changing the maximum alters the contract instead of repairing its comparison.

Repair a missing building block

Review Test Diverse Users (ai-2/34). Goal: Evaluate across varied cases. Short practice: Can these records establish performance for different user groups?. Return to this week and explain what you changed.

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain

what you changed.

Review Evidence and Sources (ai-2/29). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.